How would an economist solve climate change?

ECON-4210
Economics of Climate Change
Class 16

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Georgia Institute of Technology

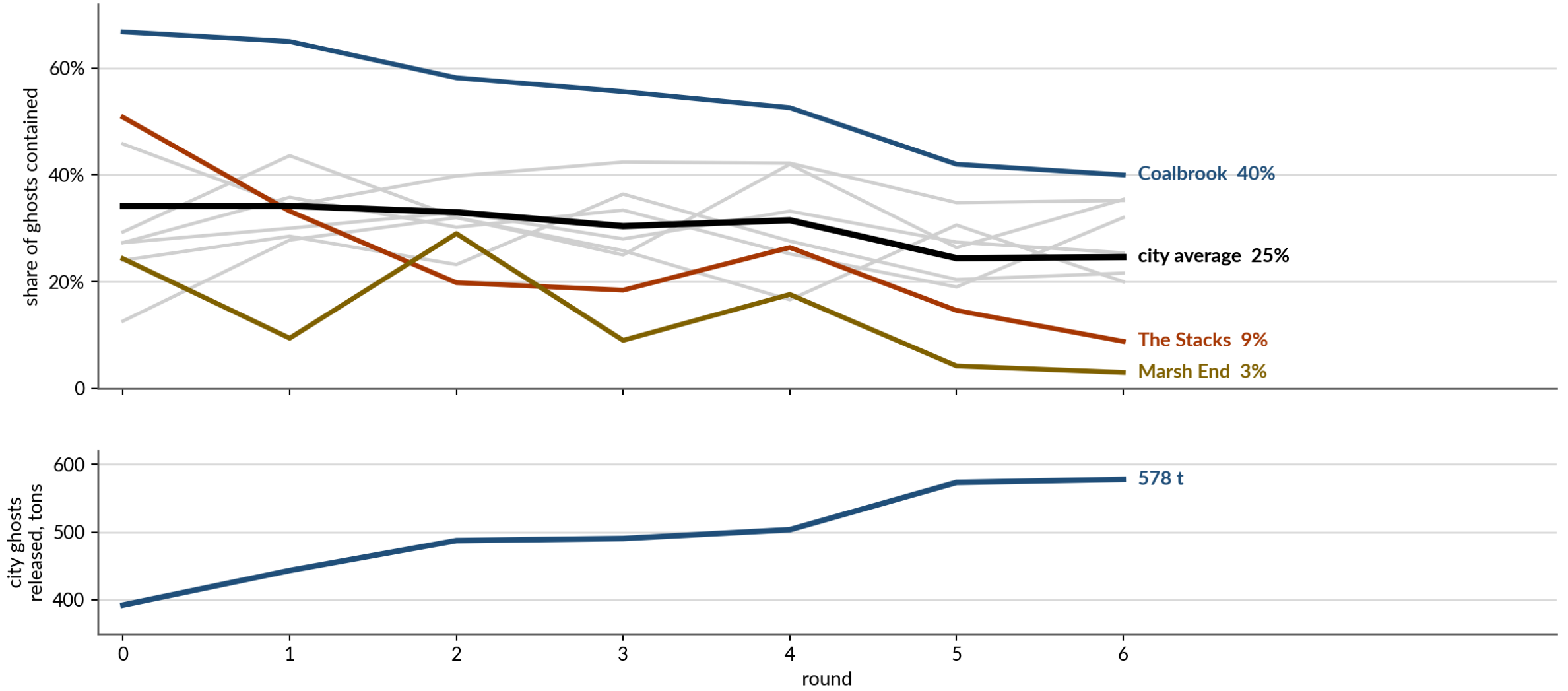
Today's class

1. Questions?
2. Attendance
3. Housekeeping
    1. Ghost Emissions: round 6 results, round 7
    2. Exam 2
4. **How would an economist solve climate change?**
    1. Cap-and-trade
5. For next class…

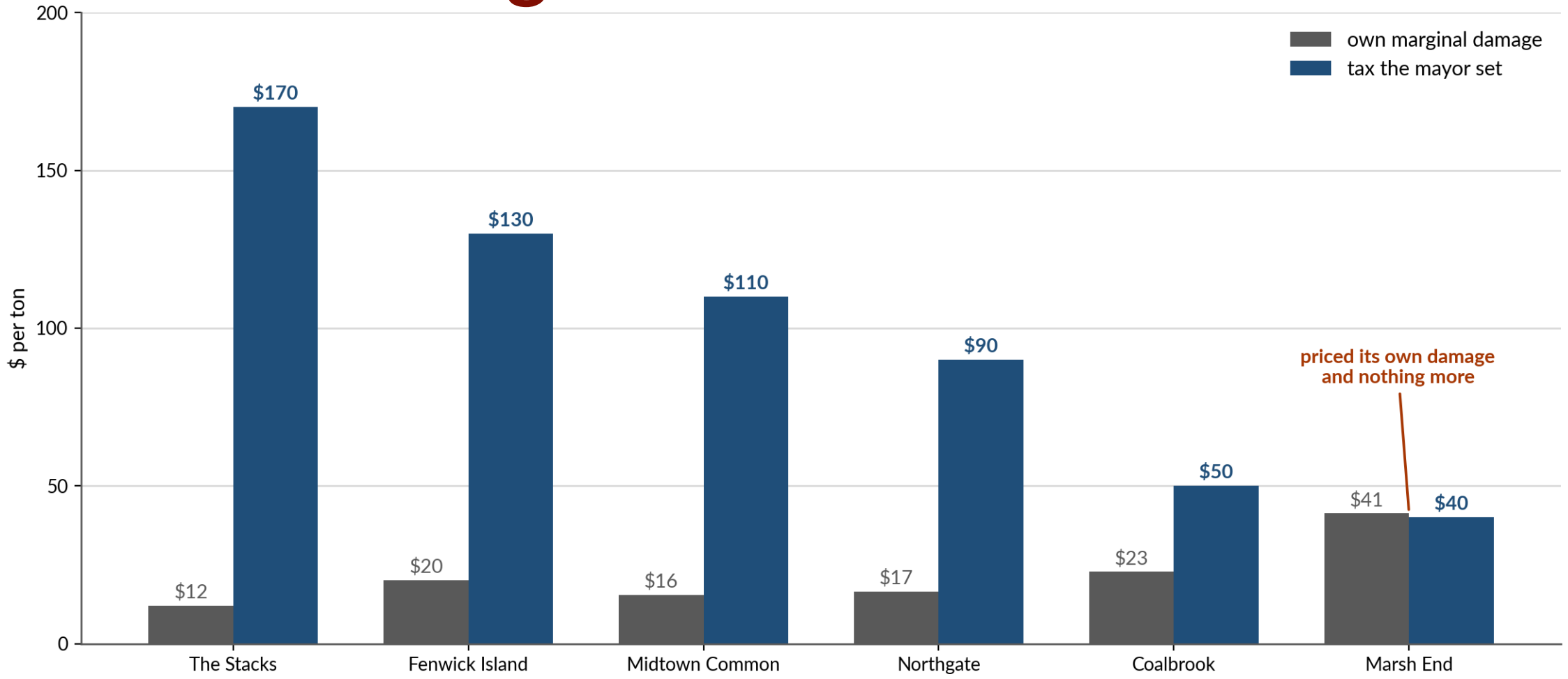
Ghost Emissions, round 6: Results

- **Tools chosen:**
  - **6 ghost taxes,**
  - **1 cap with tradeable permits,**
  - **1 minimum-containment rule.**
  - **One borough did not submit → defaulted to a ghost tax.**
- **Prices on ghosts: $40 to $170 per ton.**
  - **The city-wide SCC is $200.**
- Ghost emissions were flat, 573 to 578 tons.
  - No policy was in force yet.
  - The tools chosen above bind in round 7.
- **Question to consider:** Why did every borough price ghosts below $200?
- **Question to consider:** A tax and a cap with free permits can produce the level of abatement. Who ends up with the money under each?

Containment has fallen every round, with no policy in force



Six mayors set a tax. Only Marsh End stopped at its own damage



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Ghost Emissions, round 7: Cost shock

- Your borough's tool from round 6 applies to your business now.
  - Containment costs are uncertain.
  - The true cost is revealed after decisions close.

- **Businesses/Mayors** submit by Sunday, 11:59 PM.

- **Question to consider:** If containment turns out more expensive than expected, what happens to ghost emissions under a tax? Under a cap?
- **Question to consider:** Which one would you rather have gotten wrong?

Exam 2

- Grades are posted on Canvas.
  - Mean: 33.6/40.
  - Median: 35/40.
  - 19 students score 90%+, 11 students 80-90%, 14 students <80%

- PS2
  - Mean: 98.2%
  - Median: 100%

Exam 2: question 1.20

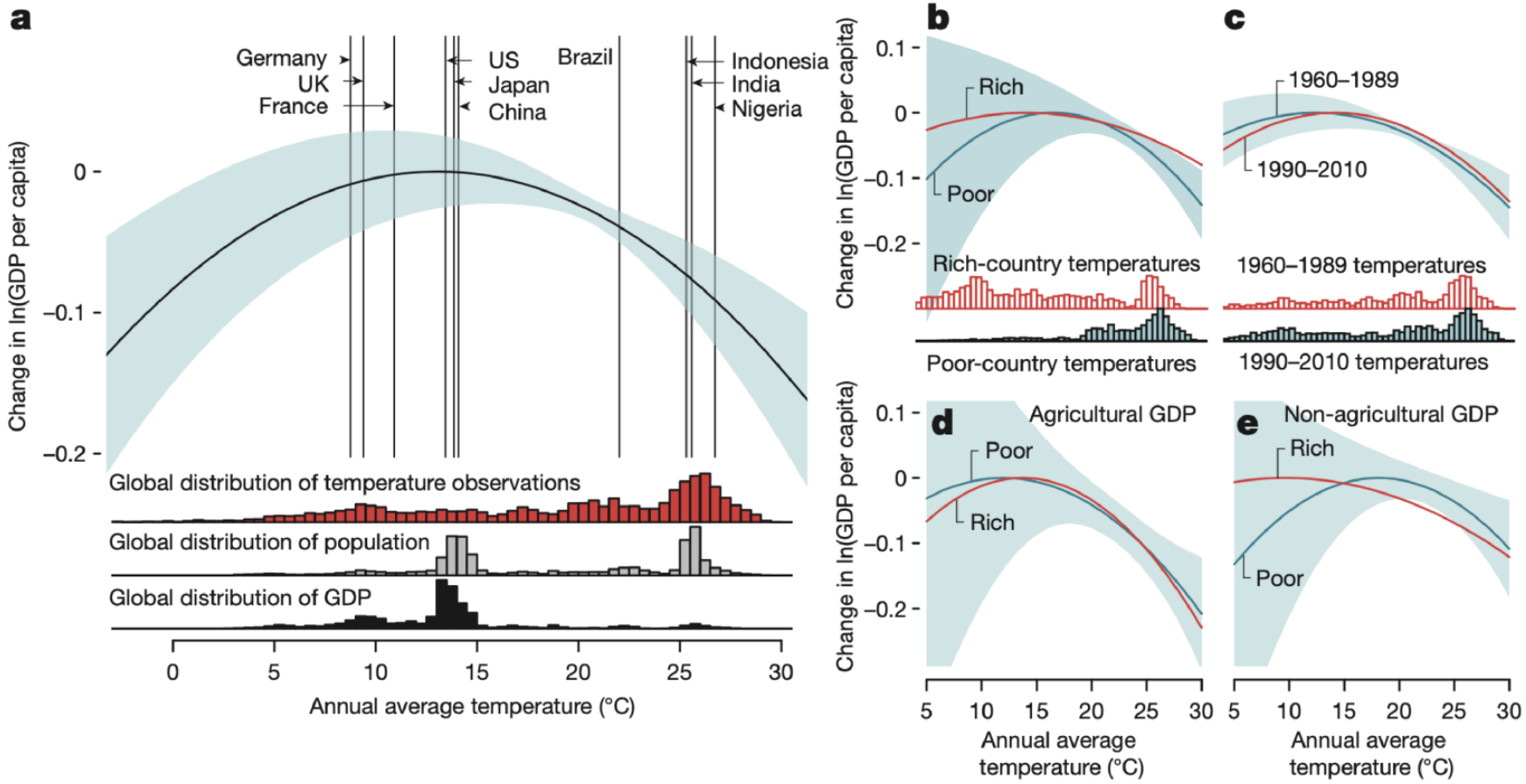


Figure 1: Main result from Burke, Hsiang, & Miguel, "Global non-linear effect of temperature on economic production," *Nature*, 2015.

Exam 2: question 1.20

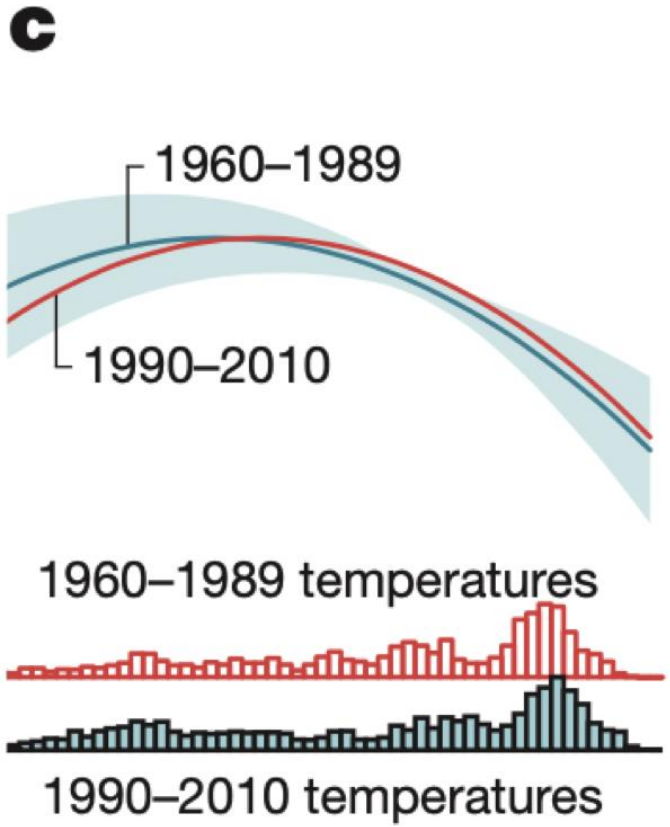
1.20. Consider the figure above from Burke et al. (2015) representing the weather-response function relating annual temperature to changes in GDP. Which of the following statements is best supported by panel (c)?

(a) The weather-response functions for both 1960–1989 and 1990–2010 are similar, which suggests that adaptation will help lessen the overall economic impact that countries will experience in the future.

(b) The weather-response functions for both 1960–1989 and 1990–2010 are similar, which suggests that poorer countries cannot "catch up" in terms of economic growth.

(c) The weather-response functions for both 1960–1989 and 1990–2010 are similar, which suggests that there is little evidence of adaptation to hotter temperatures.

(d) 1960–1989 temperatures were generally cooler than 1990–2010, so we cannot infer how climate change might affect GDP.



Exam 2: question 1.17

1.17. Rennert et al. (2022) build their estimate of the social cost of carbon from four damage sectors: agriculture, energy expenditures, sea-level rise, and heat-related mortality. Which of the following is correct?

(a) Mortality and agriculture together account for most of the estimate.

(b) All four sectors are market damages, so the estimate leaves out nonmarket damages entirely.

(c) Sea-level rise accounts for most of the estimate, because coastal property is the most valuable capital exposed to climate change.

(d) Energy expenditures account for most of the estimate, because demand for air conditioning rises steeply with temperature.

Exam 2: question 1.6

1.6. Prof. Wichman estimates how his household's daily electricity use responds to outdoor temperature, using three years of data. A linear fit and a quadratic fit give nearly identical predictions for days within the range of temperatures in the data. Under 10°F of uniform warming, the linear fit predicts a 14% increase in electricity use and the quadratic fit predicts a 37% increase. The main reason the two predictions differ is that:

(a) Many days in the warmer climate are hotter than any day in the data, so the prediction for those days depends on the shape of the fitted curve.

(b) The quadratic fit has more parameters, and a model with more parameters always predicts a larger response to any change in temperature.

(c) The linear fit builds in adaptation, such as a more efficient air conditioner in a warmer climate, and the quadratic fit does not.

(d) Three years of daily data are too few observations to estimate any relationship between outdoor temperature and household electricity use.

Exam 2: question 2.2

2.2. (10 pts) In a city of 2 million people, the number of additional deaths per 100,000 people on a hot day, compared with a mild 20°C day, is

$$M(T) = 0.8 - 0.08\,T + 0.002\,T^2,$$

where T is the day's average temperature in °C. Plug the temperature in directly: on a 30°C day, $T = 30$.

Today, the city has 30 hot days per year at 30°C. By 2100, those 30 days will be 35°C. All other days are mild. The population stays at 2 million.

(a) (3 pts) Fill in the table below. How many additional deaths per year does climate change cause by 2100?

	Today	2100
Temperature on hot days, T (°C)	30	35
Additional deaths per 100,000 per hot day, $M(T)$		
Additional deaths per year in the city		
Additional deaths per year caused by climate change		

Exam 2: question 2.2(b)

(b) (2 pts) Using a value of a statistical life (VSL) of $11 million, what is the annual dollar value of the deaths caused by climate change in 2100?

Exam 2: question 2.2(c)

(c) (3 pts) By 2100, residents install air conditioning, which flattens the response function to $M_A(T) = 0.4 - 0.04\,T + 0.001\,T^2$. Air conditioning costs residents \$45 million per year. Using $M_A(T)$, how many additional deaths per year does climate change cause in 2100? What are the total annual climate damages, including the cost of air conditioning? Compare your answer with part (b).

Exam 2: question 2.2(d)

(d) (2 pts) Suppose instead that the city is in a country with one-fifth of US income, and the VSL is scaled down in proportion to income. Recalculate your answer to part (c). Should the VSL be scaled by income in this case? Briefly make the case for the approach you prefer.

Next class…

- Wednesday, 10/28: How would an economist solve climate change? (II)
  - Read: [TBD].
- Ghost Emissions round 7 decisions due Sunday, 11:59 PM.
- In-class policy game next Monday.

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